Contextual Child Safety Risk Detector

Detecting Children and Household Hazards Using Deep Learning

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ABSTRACT

Unsupervised young children face risk of injury and harm due to harmful objects in their surroundings such as knives or tools. Currently, home monitoring systems rely on manual supervision rather than automated visual detection. This proposal presents a deep learning pipeline that detects children and hazardous objects using two separate YOLOv8 models, then combines their outputs through a proximity-based risk scoring to classify a scene as Safe or Unsafe. Expected outputs include a bounding box for both classes and a risk label, evaluated using detection accuracy (mAP) and risk classification accuracy against a manually labeled test set.

KEYWORDS

Object detection, YOLOv8, Computer Vision, Child Safety

1 Introduction

A child left alone in a kitchen for even a few minutes is enough time for a serious accident to happen, whether it be a dropped knife within reach or a small object that poses a choking hazard. This reasoning has popularized the use of home appliances such as smart home cameras and baby monitors. These smart devices have grown over the years, with examples such as CuboAI, a commercial baby monitor that uses computer vision in order to detect if the babies face is covered or if they rolled over, as well as a danger zone feature that alerts the user when a child enters a specific pre-marked area of the home such as near a staircase or pool. However, these zone-based systems are manually configured. It has no understanding of what hazard is actually present or whether one has moved since. A hazard in any predefined zone would go undetected because the system was never designed to recognize hazardous objects at all, only to check a child's coordinates against a fixed region. This project addresses that specific limitation by creating a system that can detect the hazardous object itself and compute risk based on the actual measured distance between child and hazard with no manual configuration and investigates if two independently trained specialist detectors match or exceed the performance of a single unified detection system and what are the performance and computational cost of such a system.

2 Related Work

Prior work on an automated child safety monitoring system has largely converged on the use of object detection, but differ in the way they approach how objects and risk are inferred and whether detection is treated as unified or separated into several specialized models.

Ahmad et. al (2025) is the closest existing system to this project that was found. The research used a YOLOv8 detector trained on a single merged dataset of a "baby" class and hazards such as pools, stairs, broken glass, and knives. Upon detecting the targets (baby and hazard) a euclidean distance is calculated in between the child and the nearby hazards in order to trigger a safety warning. This approach shows that YOLOv8 combined with distance based risk levels is a viable approach to the problem at hand. However, their architecture trains child and hazard detection jointly as one whole multi-class model. Meaning the child class with its own categories share model capacity and training signal with several other hazard classes of differing visual complexities. This project aims to synthesize their detection-plus distance framework but test whether using separate detection models for hazard and child detection would merit a more maintainable and higher-performing alternative against their reported 90%mAP.

Another research study conducted by AlMhdawi et. al (2026), provides a similar methodological approach but in a different domain. The study proposed a dual-based YOLOv8 framework for fire detection and proximity-aware risk assessment; these two models are fused via centroid Euclidean distance converted to real-world units and applies a exponential decay function to a risk tier (Low through Critical). This is almost similar to our proposed pipeline, differing mainly in that one of their two detectors is a general/COCO-pretrained model rather than two purpose trained specialists. Synthesized alongside Ahmad's work the two of these papers look at a central question that this project aims to investigate: does a fully independent and dual-specialized detector design approach match or exceed the performance of a single unified model (Ahmad et al.) or a hybrid of specialist plus general design (AlMhdawi et al.)

Solovyev et al. (2021) justify treating two independently trained detectors rather than a single unified model. Their method demonstrated that combining bounding-box predictions from separately trained models can match or exceed the performance of a single trained detector without requiring joint training. This directly supports this project's decision to keep the child detector and hazard detector separate rather than merging their training data. This shows that fusion at the output level rather than at the training level is a proven and sufficient strategy for combining detectors.

Khan and Dey (2024) approached the problem of child safety from a different perspective by focusing on the classification of safe and unsafe household objects rather than on real-time surveillance. They introduced the ChildSUn (Child Safe and Unsafe) dataset which consists of approximately 5350 images across ten categories of safe and unsafe. Unlike our proposed project, they only focused on image classification and did not further explore other aspects such as the spatial relationship between the child and the object.

Ramadan et al. (2025) proposed a computer vision based child safety monitoring system which combines YOLOv11n with human pose estimation. The goal of this was to detect situations where the child may place objects in their mouth. Similar to other studies, the system also uses the euclidean distance. However, the distance being measured is between the child's hand and mouth, rather than the distance of an object and the child. The study reported an accuracy of 92% in scenarios without objects and only 74% with objects. It shows that doing two recognition tasks with one detector may have a lower accuracy compared to the proposed study which separates it by using two independent trained detectors..

3 Methodology

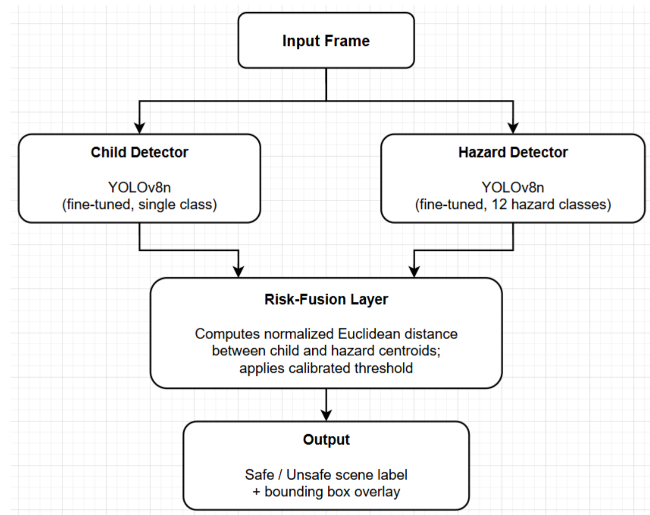

Figure 1: **Proposed architecture for contextual child safety risk detection**

1. **Input:** a single image frame, or one frame extracted from a live camera feed.
2. **Child detector:** a YOLOv8n model fine-tuned exclusively on the child-detection dataset (1,985 images, single class). Outputs bounding boxes for the child class, each with a confidence score.
3. **Hazard detector:** a separate YOLOv8n model fine-tuned exclusively on the harmful-objects dataset (12 classes: Axe, Chainsaw, Chisel, Coin, Drink, Dumbbell, Fork, Screwdriver, Stapler, Knife, Hammer, Scissors). All 12 classes are collapsed into a single hazard label at the fusion stage, though the original per-class label is retained in the output for interpretability (e.g., displaying "Knife" rather than just "hazard").
4. Both detectors run independently on the same input frame, with no shared backbone or joint training, and each is trained, validated, and tested entirely on its own dataset.
5. Risk-fusion layer:
    a. If either detector returns zero detections (no child, or no hazard) classify the scene as Safe.
    b. If both return at least one detection, compute the Euclidean distance between

every detected child box center and every detected hazard box center:

$$d_{norm} = \frac{euclidean\ distance(child, hazard)}{image\ diagonal}$$

- Take the minimum distance across all child-hazard pairs in the frame.
- If *min_distance_norm* falls below a calibrated threshold, label as unsafe; otherwise, label as safe.

6. **Output:** the original frame annotated with both detectors' bounding boxes, plus an overlaid Safe/Unsafe label.

This design keeps two detection tasks fully separated rather than a single model across both datasets. The child detector and hazard detector are each optimized independently on data drawn from its own domain. The risk-fusion layer uses a distance threshold instead of a trained model. This choice is grounded in Weighted Boxes Fusion (Solovyev et al., 2021), which demonstrates that combining outputs from independently trained detectors is a proven strategy that avoids the class imbalance and shared-capacity issues a single merged model would face.

4 Experimental Plan

These models would be trained, validated, and tested through the ultralytics library. This library is built on top of PyTorch, and it focuses on developing and maintaining models under Ultralytics YOLO (You Only Look Once), which includes the YOLOv8 models we will be training. These models will be trained on Google Colab, using a T4 GPU. The estimated training time for these models would depend on the value set for epochs. For a standard value of 100 epochs, it can take several hours.

Two computer vision datasets from Roboflex Universe will be used in conjunction: a child detection dataset and harmful object dataset. We will train one model on the child detection dataset and another on the harmful object dataset.

The child detection dataset is filled with 5917 images of children that are annotated to specify where the children are located in an image (if a child is present). The creator of this dataset has created their own train-test-val split: 87% train, 8% validation, and 5% test.

The harmful object dataset is filled with 4705 harmful objects that pose risk, especially to those under 14 years old. These photos are annotated similarly, specifying where these dangerous objects are (if they are present). The creator of this dataset has created their own train-test-val split: 80% train, 13% validation, and 7% test, so we will be utilizing this split.

As mentioned in the methodology, the data will be used to train two separate YOLOv8 nano (YOLOv8n) models. For both models, we would be optimizing the following parameters through hyperparameter tuning: initial learning rate (lr0), box, cls, and dfl. The latter 3 of these parameters are the loss weighting parameters, meaning that their values affect how much weight certain aspects in the model have in the output, namely, the accuracy of the box highlight and the accuracy of the classification.

The YOLOv8 large model trained by Ahmad et al. (2025) will be one of the main baselines we will utilize for our study. Since it runs a similar pipeline of detection then distance to assess danger, it would be a suitable comparison to our pipeline.

There are several ablation studies that we will be performing. The first two are primarily for the setup of the models: input resolution comparisons and optimizer comparisons. These hyperparameters are not typically trained and optimized, but rather, they are decisions made by the researchers that would be best for the object detection, as they tend to be consistent throughout all images once it has been decided. To be specific, input resolution is denoted by the imgsz parameter in the ultralytics library, and it indicates what size and resolution of an input image is set before being fed into the network, regardless of what it was originally. Testing this will be crucial to see if the higher resolution is a necessary choice for the object detection models to improve precision. Optimizer refers to the different optimizer parameters that can be set for the model. Optimizers control how gradients are utilized, and the choices available in this parameter include several different options, such as SGD and Adam. This ablation study will be conducted to determine which of these we would be utilizing.

On top of ablation studies to determine optimal model setup, some ablation studies will be conducted to test our architecture's ability to perform the needed subtasks, namely: object detection ability and distance point references. The ablation study on object detection ability will test the ability of our models to identify and locate children and harmful objects, respectively, in comparison to our baselines. The object detection is how the risk calculation can even happen, so we would want to see if it can perform at the level of existing models that do this. The ablation study on distance point references will compare using centroids to calculate the distance of the child and the harmful object with using the bounding box to calculate it. Centroids may not be able to tell the complete story on how unsafe a situation may look like, as centroids of children and harmful objects can be slightly farther from each other while the nearest point of a child and the nearest point of the harmful object may be nearer, increasing the risk level. This ablation study will be conducted to see whether that better determines whether the situation is safe or not.

There are two key evaluation metrics we would be utilizing for our risk detector: Mean Average Precision (mAP) and Euclidean distance. mAP works by calculating the average precision per decision class, then afterwards, you calculate the mean of each of those average precision values. This would be utilized primarily for the hazard detector in measuring the precision of labeling the correct class among the 12 different hazard classes. Euclidean distance is a formula that calculates the distance of two objects using the following formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

where x_1, y_1 are coordinates of one centroid, while x_2, y_2 are coordinates of another centroid. This formula will be utilized in the Risk-Fusion Layer, wherein the centroids of the boxed children and the centroids of the dangerous objects will be placed in the Euclidean distance formula to measure whether or not the children and the harmful objects are within the threshold of an unsafe distance of 700 pixels (Ahmad et al., 2025)

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